Milestone 2 Report: Core Data Analysis and Visualization

Project Title: ClimateScope – Global Weather Insights

Student Name: Srija

Milestone: 2

1. Introduction

This milestone focuses on exploratory data analysis and visualization using the cleaned weather dataset from Milestone 1. The main aim is to understand patterns, trends, correlations, and extreme weather events. This analysis helps in preparing the data for predictive modeling in later stages.

2. Dataset Overview

The dataset contains global weather observations collected from various locations. It includes meteorological data, geographical information, and air quality indicators along with time-related attributes.

Key attributes used:

- Temperature (°C)

- Precipitation (mm)

- Humidity

- Wind speed and direction

- Pressure

- Cloud cover

- Air quality parameters (PM2.5, PM10, CO, NO2, O3, SO2)

- Latitude and longitude

- Time of observation (last_updated)

3. Statistical Analysis

Descriptive statistics were calculated to understand the central tendency and variability of the data. Temperature and air quality parameters showed significant variation across regions. Precipitation values were mostly low with occasional extreme values.

4. Distribution Analysis

Histograms were used to study the distribution of key variables.

Temperature showed a near-normal distribution, while precipitation was right-skewed, indicating rare heavy rainfall events.

5. Correlation Analysis

Correlation analysis was performed on numeric attributes using a correlation matrix and heatmap. Strong correlation was observed between temperature and feels-like temperature. Air quality indicators were also interrelated, while precipitation showed weak correlation with temperature.

6. Trend and Seasonal Analysis

Time-based analysis was done using the last_updated column. Daily and monthly aggregations helped identify temperature trends and seasonal variations. Clear seasonal patterns were observed across months.

7. Extreme Weather Event Detection

Extreme weather events were identified using percentile-based thresholds. The top 5% of temperature values were considered extreme heat events, and the top 5% of precipitation values were considered extreme rainfall events.

8. Regional and Spatial Analysis

Country-wise aggregation was performed to compare average temperatures across regions. Geographic visualization using latitude and longitude highlighted global temperature distribution and regional climate differences.

9. Visualization Design

Appropriate visualization techniques were selected based on analysis needs:

- Line charts for trends
- Histograms for distribution
- Heatmaps for correlation
- Bar charts for regional comparison
- Geographic plots for global patterns

10. Conclusion

Milestone 2 successfully analyzed the cleaned weather dataset and identified meaningful insights related to trends, correlations, seasonal patterns, and extreme events. This milestone provides a strong foundation for building predictive models in the next phase.